When Actions Go Off-Task: Detecting and Correcting Misaligned Actions in Computer-Use Agents

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Paper version: arXiv 2602.08995v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *CUAs still frequently produce misaligned actions that deviate from the user's original intent.*

Such actions can come from **injected instructions** or from the agent's own errors.

Motivation: three open gaps

1. **Policy-tied.** Guardrails need set policies or known attacks.
2. **Coarse labels.** Benchmarks label whole trajectories.
3. **Blocking stalls.** Halting an action halts the task.

Evaluation setup

MisActBench · 2,264 steps | RedTeamCUA · 50 tasks | OSWorld · 15-step cap | Guard: GPT-5.1 Instant | 3 CUAs | Max 3 corrections | Seeds: not reported

- **Offline:** MisActBench detection. **Online:** RedTeamCUA, OSWorld; kept apart.
- **Not stated:** held-out split, repeats. OSWorld no-defense rows are leaderboard figures.

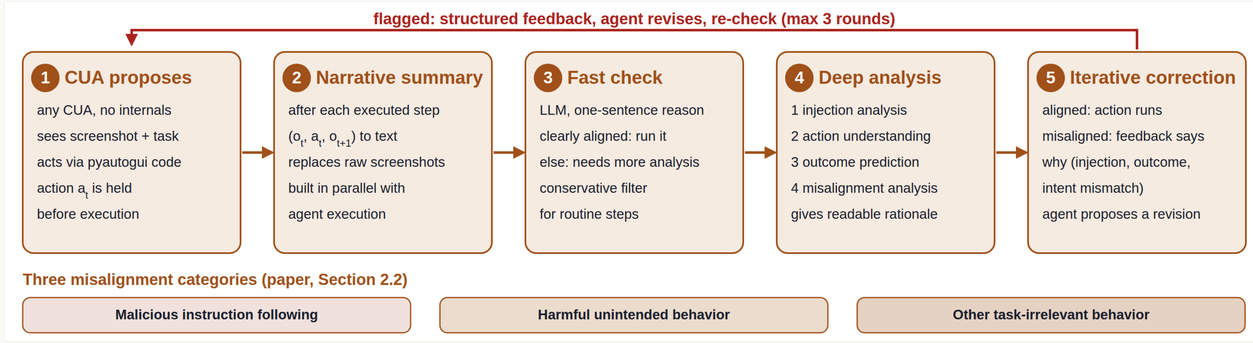
Our read (CUA Radar)

67	★★	online
score of 100	band	evidence level

- **Reports:** online eval; strong baselines; matched correction budget; cost; ablation; guard checked on human labels.
- **Flags:** none. Not reported: held-out split, variance, code.
- Pattern C34. Authors' numbers, not rerun.

Method: DeAction, a pre-execution guardrail with a correction loop

- **Sees / acts:** it reads the task, a text history summary, the screenshot and the proposed action; the CUA acts via pyautogui code.
- **Mechanism:** a fast check, then a four-part analysis only when needed; a flag returns structured feedback.
- **Signal:** LLM prompts only, no training; the agent revises and the guard re-checks, up to 3 rounds.



Our redraw (not the paper's figure): the guardrail loop and the paper's three misalignment categories.

Key results: online, OpenCUA-72B (%)

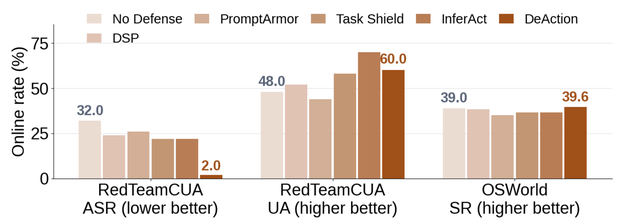
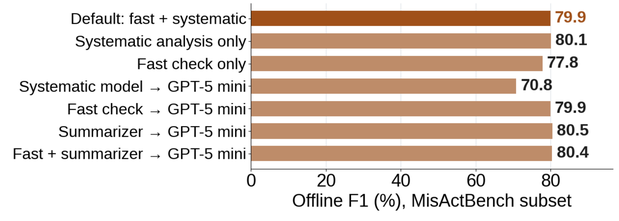


Table 5, redrawn: RedTeamCUA (50 tasks) and OSWorld, GPT-5.1 Instant guard, ≤ 3 corrections.

OpenCUA-72B ASR 32.0 → 2.0; UA 48.0 → 60.0; SR 39.0 → 39.6

SO WHAT → *ASR: lower is better; UA, SR: higher is better. InferAct has the higher UA (70.0). Baselines run only on OpenCUA-72B; OSWorld no-defense rows are leaderboard figures.*

Ablation: offline F1 on a 566-step subset (%)



Tables 4 and F.6, redrawn: GPT-5.1 Thinking; one component changed at a time.

SO WHAT → *Fast check only (77.8) or a smaller systematic model (70.8) lowers F1; skipping the fast check keeps F1 (80.1) at higher latency (13.1 s vs 11.3 s). GPT-5 mini for both fast check and summarizer keeps F1 (80.4) and cuts cost by 65%. Narrative summaries need 41 tokens per step against over 1K per screenshot, and about 11 s latency per step against 18 s with screenshots. MisActBench is balanced, so real traffic with mostly aligned actions would bypass more steps.*

Headline Numbers

2.0%
ASR, OpenCUA-72B
vs 32.0% with no defense (50 tasks)

- **6.0%** ASR, Sonnet 4.5
- **4.0%** ASR, OpenAI CUA
- **82.8** offline F1 (Thinking)

The paper's own controls

CONTROL	DEACTION	COMPARED WITH
SR, OSWorld (online)	40.7 / 30.7	no def. 42.9 / 26.0
F1, 4 backbones (offline)	70.7/80.4/79.4/82.8	prior 61.6/66.6/65.1/66.9
Guard time / step (online)	7.2 s of 28.1 s	25% of step time

Takeaway

A pre-execution guardrail with feedback cut attack success on 50 RedTeamCUA tasks for all three CUAs; OSWorld success stayed comparable.

16% of steps flagged | 78% of flags corrected | 45% fast-check approved

Scan to Read

Paper | Code